Clinical review of robotic distal pancreatectomy

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IHPBA 2026 SINGAPORE 28–31 OCTOBER 2026

Robotic distal pancreatectomy (RDP) matches or improves on laparoscopic surgery and outperforms open surgery, at a similar overall cost to laparoscopy.

≈1 day	0%	≤ LDP	+30 min
shorter hospital stay than laparoscopic	positive margins (laparoscopic 2%)	overall cost; well below open surgery	operating time vs laparoscopic, mostly docking

01 Background & aim

- Distal pancreatectomy has moved from open (ODP) to minimally invasive surgery: laparoscopic (LDP) and robotic (RDP).
- LDP is established but limited by dexterity and visualisation.
- Whether RDP's technical gains translate into better outcomes is still debated.

Aim Compare RDP with ODP and LDP: technique, outcomes, cost-effectiveness and challenges.

02 Why robotic?

3D magnified visualisation

Tremor filtration

Motion scaling

Wristed instruments

Platforms: almost all evidence is from da Vinci; Versius and hinotori™ have only early case reports.

03 Operative technique

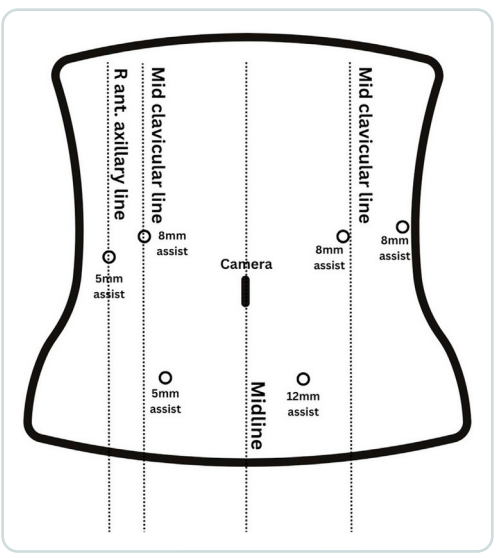


Fig. 1 Port placement

1. **Position** Supine; reverse Trendelenburg; dock
2. **Expose** Enter the lesser sac
3. **Control** Splenic artery and vein
4. **Divide** Pancreas, clear margin
5. **Resect** En bloc ± spleen

Spleen preserving: splenic vessel preservation or Warshaw technique

04 Clinical outcomes of RDP

Outcome	RDP vs open	RDP vs laparoscopic
Blood loss	▼ Lower	▼ Lower (planned SP)
Complications	▼ Fewer (fistula, wound)	= Similar overall and fistula
Hospital stay	▼ Shorter	▼ ≈1 day shorter
Conversion	Not applicable	▼ Fewer to open
Spleen preservation	▲ Higher	▲ Higher
Oncological	= Non-inferior R0 and nodes	▲ 0% vs 2% R1; more nodes
Operative time	= Conflicting	▲ ≈30 min longer

■ favours RDP ■ no clear difference ■ favours comparator SP, spleen preservation

05 Cost-effectiveness

Operative cost	Overall cost	Quality of life
RDP higher	RDP ≤ LDP < ODP	RDP > LDP

- Theatre costs are offset by shorter stay, fewer conversions and lower morbidity.
- Costs fall as case volume grows.
- More QALYs than LDP at similar cost (ROBOCOSTES and others).

06 Challenges

Operative time
Docking adds time; falls along the learning curve

Team & training
Dedicated robotic team; few training centres

Access
High upfront cost; robots concentrated in advanced economies

CONCLUSION

RDP offers greater precision, better clinical outcomes and more spleen preservation than open and laparoscopic surgery, and is cost-effective. Wider adoption depends on training and equitable access.

Full paper: Lap Endosc Robot Surg 2025;8:80–83